MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 8943

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Coaxial fixed frequency, 16.5 GHz, integral magnet, pulsed, 65 kw minimum peak power output.

ABSOLUTE RATINGS:

Parameter:	Ef	ib	Pi	pi	tpc	rrv	Anode T	Bushing T	Altitude
Unit:	V	a	W	kw	µsec	kv/µsec	°C	°C	feet
Maximum:	14.0	17	210	305	1.2	180	160	170	70.000
Minimum:	----	--	---	---	---	---	-62	-62	----

PHYSICAL CHARACTERISTICS:

Dimensions:	See Figure 2	Cooling:	Forced air (See note 9)
Mounting position:	Any		
Coupling:	Output flange compatible with UG541A/U	Weight:	5.75 pounds (maximum)

TEST CONDITIONS:

Test Frequency: 16,500 ±35 MHz

Parameter:	Ef	If	tk	tpc	prf	rrv	Du	Ib	VSWR
Unit:	Vac (See note 1)	A	Sec	µs	pps	kv/µs (See note 4)	---	mAdc	----
TEST 1									
Maximum:	7.0	2.0	180	0.24	4500	180	---	----	1.5
Minimum:	6.0	---	---	0.16	3240	140	.00072	11.5	---
TEST 2									
Maximum:	10.0	2.0	180	1.2	720	180	---	----	1.5
Minimum:	9.0	---	---	0.8	648	140	.00072	11.5	---
TEST 3									
Maximum:	13.0	2.5	180	0.24	720	180	---	----	1.5
Minimum:	12.0	---	---	0.16	648	140	.00014	2.31	---

GENERAL:

Qualification - Required

Replaces TI Part Number 516994-1
Varian Type SFD-342
Litton Type L4419

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection						
1011	Humidity	1	MIL-STD-202, Method 103A Test condition B	---	---	---	---
----	Barometric pressure (altitude)	1	MIL-STD-202, Method 105C Test condition C	---	---	---	---
----	Temperature cycling	1	MIL-STD-202, Method 102A Test condition C				
	Quality conformance inspection, part 1						
4289	Heater current		E_f = 12.5 ±0.5 Vac	If	1.8	2.5	A
4306	Pulse voltage	1		epy	14.0	18	kV
4250	Power output	1		Po	46.8	61.2	W
4308	R F Bandwidth (1)	1		BW	----	$\frac{1.9}{tpc}$	MHz
4308	Minor lobe ratio(1)	1		Ratio	9	---	db
4310	Frequency pulling factor	1		ΔF	----	15	MHz
4308	R F Bandwidth (2)	2		BW	----	$\frac{1.9}{tpc}$	MHz
----	Frequency	3		F	16,465	16,535	MHz
4308	Minor lobe ratio(2)	2		Ratio	9	----	db
4315	Pulse stability	2	See note 5	MP	----	.25	%
	Quality conformance inspection, part 2						
4003	Pressurizing	1	See note 3; 40 lb_f/in^2 input and output assemblies	---	----	----	---
4308	R F Bandwidth(3)	3		BW	---	$\frac{1.9}{tpc}$	MHz
	Quality conformance inspection, part 3						
----	Life test	2	Group D; See note 11	t	500	---	hrs
	Life test end points:						
4250	Power output	2		Po	43	--	W
4315	Pulse stability	2		MP	---	0.5	%
1032	Vibration, mechanical	2	See note 7	ΔF	---	0.5	MHz
1042	Shock	2	See note 8	---	---	---	---
4027	Temperature	2	See note 2	ΔF/ΔT	---	0.4	MHz/°C

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NOTES:

1. Prior to the application of high voltage, the cathode shall be heated to the required initial operating temperature. This may be done by applying 12.5 volts ±5 percent for three minutes. On the application of anode power, the heater voltage should be lowered to the voltage specified, and for various power inputs, it should be adjusted within ±0.5 VAC, according to the following formula:

   $$Ef = 12.5 - 0.0232\ (Pi)$$

   The tube heater may be protected against arcing by use of a connector that places a minimum capacitance of 4,000 pf across the heater directly at the input terminals.

2. The temperature is to be measured at the point indicated on the outline drawing.

3. The mechanical design of the mounting plate shall provide for cathode terminal pressurization of 25 ±2 lb_f/in^2 gage and leak rate not to exceed 0.1 standard cubic inch per minute. The surface of the output flange shall be capable of pressurization with an "O" ring in the mating flange.

   The cathode terminals shall operate in air pressurized to at least 12 lb_f/in^2. The output window shall operate in air pressurized to at least 20 lb_f/in^2 when the output mismatch is higher than 1.3 VSWR or 15 lb_f/in^2 when the output mismatch is less than 1.3 VSWR.

   The magnetron may receive forced air cooling, to keep it within the temperature rating during test.

4. The rate of rise of voltage (rrv) shall be expressed in kilovolts (kv) per microsecond (μsec) defined by the steepest tangent to the leading edge of the voltage pulse above 50 percent amplitude. Any capacitance used in viewing shall not exceed 6.0 pf. The minimum value for test shall be the maximum value of rrv for equipment use.

5. A missing RF pulse is defined as an RF pulse whose energy content (integrated power) within a ±1% frequency range of the normal operating frequency is 70% or less than that of a normal pulse. The measurement shall be made during any three minute interval of a six minute test period.

6. The magnetron will be operated both intermittently and continuously at relative humidities up to and including 100% at temperatures up to 71°C including those conditions wherein condensation takes place on magnetron.

7. The magnetron shall be supported on normal mount plate, and subjected to the specified vibration in each of the three (3) orthogonal axes. The performance of the magnetron shall be monitored and the points of maximum deviation of performance and/or resonances shall be individually tested for a period of 30 minutes each. The total vibration time in each axis shall be three hours. The total cycling time will depend upon the number of resonances, but will not be less than one hour. The maximum ΔF measured at the resonances shall be 0.5 MHz.

8. The magnetron shall be capable of specified performance subsequent to the following shock tests.

   Under normal conditions, the magnetron will be subjected to 18 impact shocks of 15 g each, consisting of three shocks each in opposite directions along each of three (3) orthogonal axes, each shock impulse being a half sine wave of 11 ±1 milliseconds time duration. The g value shall be within ±10 percent when measured with a 250 Hz low pass filter and the maximum g shall occur at approximately 5.5 milliseconds.

   The magnetron under normal mounting conditions shall be subjected to 12 impact shocks of 30 g, consisting of two shocks in opposite directions along each of three (3) orthogonal axes. Each shock impulse shall have a time duration of 11 ±1 milliseconds and the g value shall be within ±10 percent when measured with a 250 Hz low pass filter. The maximum g value shall occur at approximately 5.5 milliseconds.

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NOTES: -Continued

9. With the magnetron properly mounted on a suitable heat sink and with 20 ft^3/min air flowing around the body, the peak temperature shall not exceed the maximum rating. The anode temperature shall be measured at that point on the anode block specified in the outline drawing. The input terminal temperature shall be determined at that point specified on the outline drawing.

10. Reclaimed materials shall be utilized to the maximum extent possible.

11. Cycle tests of 30 minutes ON preceded by three minute maximum filament schedule and followed by 27 minutes OFF shall be made until the total accumulated ON time equals the specified life.

During the ON periods, a VSWR of 1.5:1 maximum shall be introduced into the external load circuit and the phase automatically adjusted at least four times during each ON period to include those points of maximum and minimum power and frequency.

Cooling during on and off periods shall be adjusted so that:

a. The anode block temperature during the ON cycle approaches but does not exceed the maximum rating (however no heat source will be required to force temperature above the value resulting from zero air flow).

b. The anode block temperature at termination of the OFF cycle shall be no higher than room temperature plus 20°C (Both off time and air quantity may be adjusted).

12. Unless otherwise specified, the AQL for all tests listed under Quality Conformance Inspection, part 1 shall be 1.0 percent, inspection level II.

Custodians:

Air Force - 85
Navy - EC
Army - ER

Review activities:

Air Force - 99
DLA - ES

User activities:

Air Force - 11
Navy - OS, AS, SH, MC, CG

Preparing activity:

Air Force - 85

(Project No. 5960-3098)

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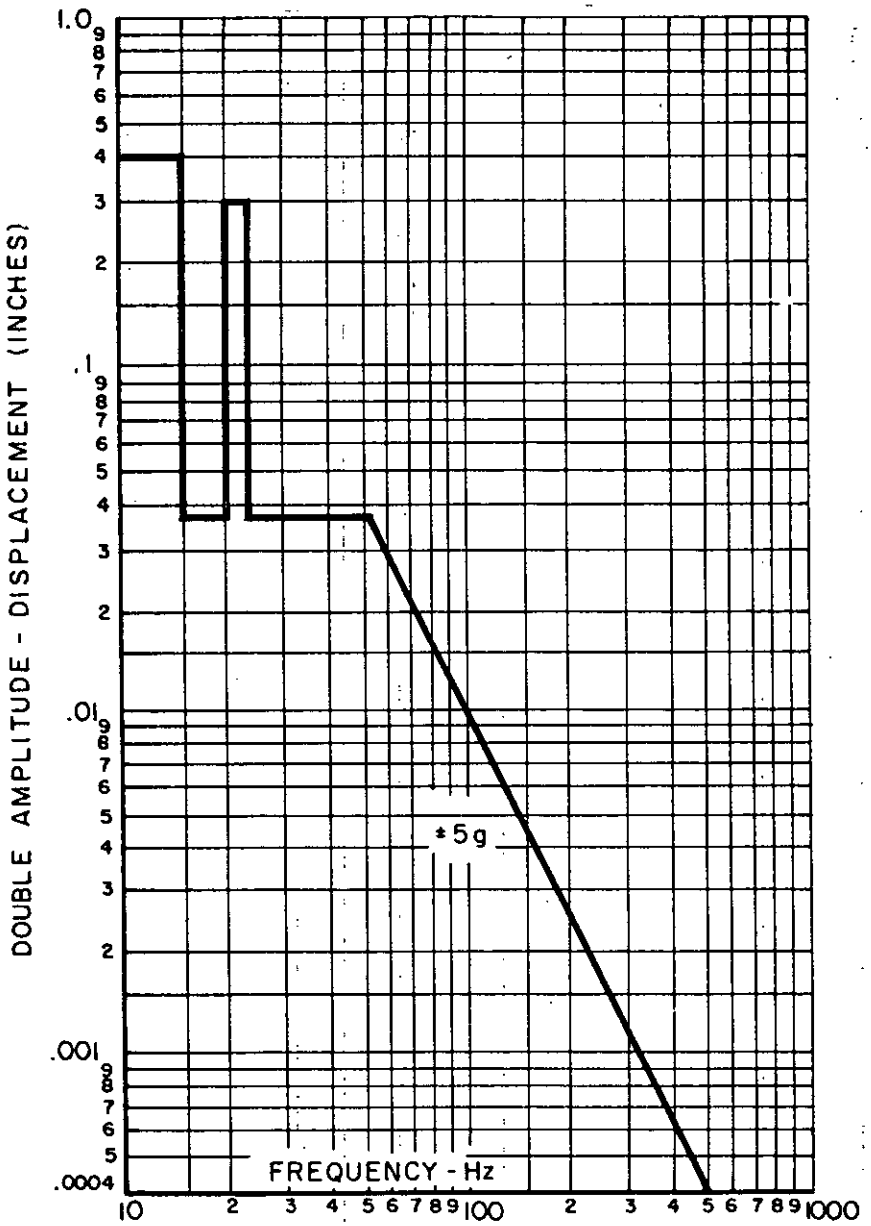


FIGURE 1 Vibration requirements

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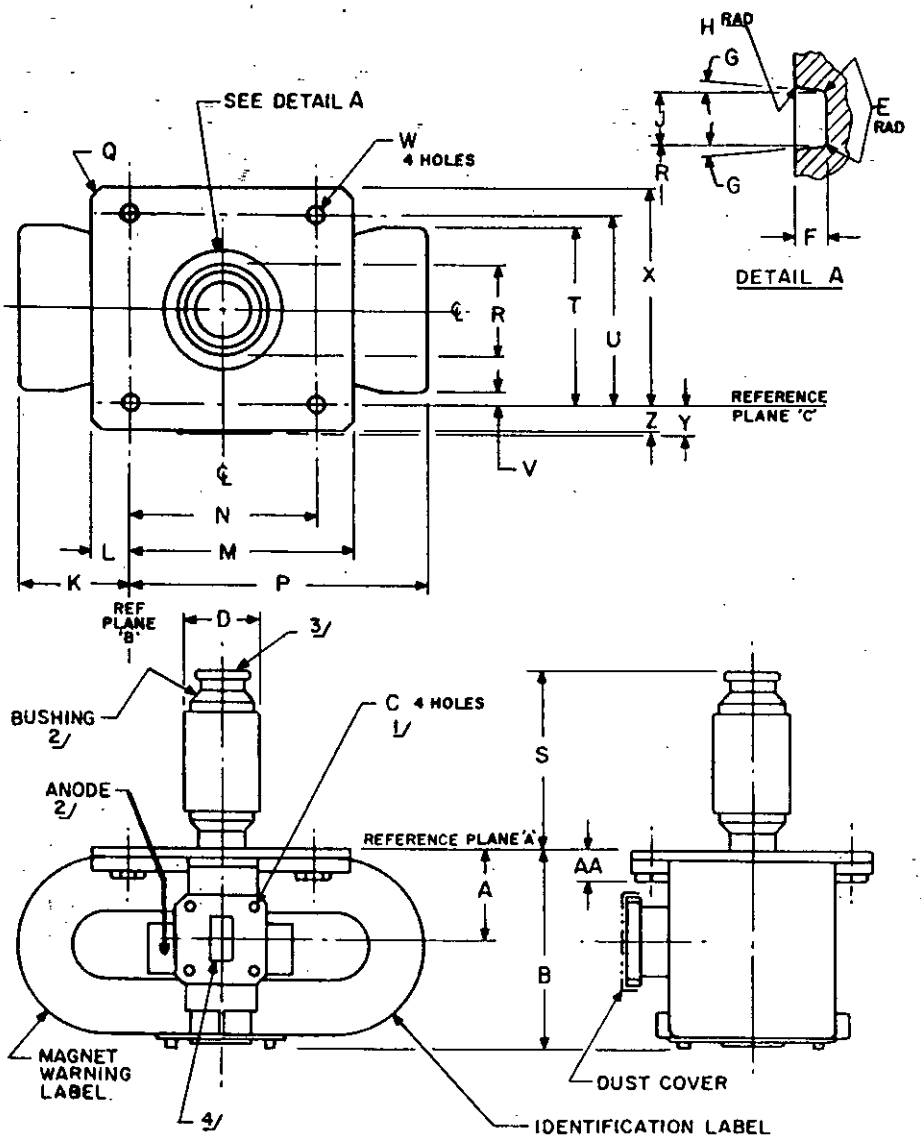


FIGURE 2 Outline of tube type 8943

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DIMENSIONS

LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART I				
A	1.22	1.42	30.99	36.07
B	–	3.0	–	76.2
C	.138-32 UNC-2B THD (4 HOLES)			
D	–	1.10	–	27.94
T	–	2.625	–	66.68
W	.297	.307	7.54	7.80
Y	.395	.455	10.04	11.56
S	–	2.625	–	66.68
AA	.455	.495	11.56	12.57
QUALITY CONFORMANCE INSPECT., PT. 3 (PERIODIC)				
E	.027	.037	.69	.94
F	.070	.090	1.78	2.29

LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
G	–	5°		
H	.005	–	.13	–
J	.185	.205	4.70	5.21
K	–	1.625	–	41.28
L	–	.545	–	13.84
M	–	3.295	–	83.69
N	2.740	2.760	69.50	70.10
P	–	4.375	–	111.13
Q	–	.125x45°	–	3.18x45°
R	1.343	1.363	34.11	34.62
U	2.815	2.835	71.50	72.01
V	.200	–	5.08	–
X	–	3.192	–	81.08
Z	–	.365	–	9.27

NOTES:

1/ MATE WITH UG-541A/U WAVE-GUIDE

2/ MEASURE TEMPERATURES AT THESE POINTS.

3/ MATES WITH CONNECTOR TI 518609 (JETTRON 9000-C) OR EQUAL.

4/ CERAMIC TYPE OUTPUT WINDOW.

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